

Kafka Setup

<https://kafka.apache.org/quickstart/>

Download: https://www.apache.org/dyn/closer.cgi?path=/kafka/4.2.0/kafka_2.13-4.2.0.tgz

Extract and rename folders to Kafka

if possible remove the second Kafka folder and move contents to the first folder

Open Powershell go to the Kafka directory

```
$KAFKA_CLUSTER_ID = .\bin\windows\kafka-storage.bat random-uuid
```

Write-Output \$KAFKA_CLUSTER_ID

```
f8CoHH7JSqCsDZkkTxMqTQ
```

```
.\bin\windows\kafka-storage.bat format --standalone -t $KAFKA_CLUSTER_ID -c .  
\config\server.properties
```

```
Works: bin\windows\kafka-storage.bat format -t YmcxyIfbTJChinnxqMf34w -c  
config\server.properties
```

In case of error:

```
2026-05-26T19:08:29.261447Z main ERROR Reconfiguration failed: No configuration found for  
'764c12b6' at 'null' in 'null'
```

Log directory /tmp/kraft-combined-logs is already formatted. Use --ignore-formatted to ignore this directory and format the others.

Delete /tmp/kraft-combined-logs folder

```
PS C:\Users\saito\Downloads\kafka> bin\windows\kafka-storage.bat format -t
```

```
YmcxyIfbTJChinnxqMf34w -c config\server.properties
```

```
2026-05-26T19:10:58.468210600Z main ERROR Reconfiguration failed: No configuration found for  
'764c12b6' at 'null' in 'null'
```

Bootstrap metadata:

```
BootstrapMetadata(records=[ApiMessageAndVersion(FutureLevelRecord(name='metadata.version',  
featureLevel=29) at version 0),
```

```
ApiMessageAndVersion(FutureLevelRecord(name='eligible.leader.replicas.version', featureLevel=1)  
at version 0), ApiMessageAndVersion(FutureLevelRecord(name='group.version', featureLevel=1) at  
version 0), ApiMessageAndVersion(FutureLevelRecord(name='share.version', featureLevel=1) at  
version 0), ApiMessageAndVersion(FutureLevelRecord(name='streams.version', featureLevel=1) at  
version 0), ApiMessageAndVersion(FutureLevelRecord(name='transaction.version', featureLevel=2)  
at version 0)], metadataVersionLevel=29, source=format command)
```

Formatting metadata directory /tmp/kraft-combined-logs with metadata.version 4.2-IV1.

Works! bin\windows\kafka-server-start.bat config\server.properties

bin\windows\kafka-topics.bat --list --bootstrap-server localhost:9092 from CMD works!

I ran in Powershell

```
transaction.timeout.expiration.ms = 604800000
unclean.leader.election.enable = false
unclean.leader.election.interval.ms = 300000
unstable.api.versions.enable = false
unstable.feature.versions.enable = false
(org.apache.kafka.common.config.AbstractConfig)
[2026-05-26 15:34:24,491] INFO [BrokerServer id=1] Waiting for the broker to be unfenced (kafka.server.BrokerServer)
[2026-05-26 15:34:24,514] INFO [BrokerLifecycleManager id=1] The broker has been unfenced. Transitioning from RECOVERY to RUNNING. (kafka.server.BrokerLifecycleManager)
[2026-05-26 15:34:24,515] INFO [BrokerServer id=1] Finished waiting for the broker to be unfenced (kafka.server.BrokerServer)
[2026-05-26 15:34:24,516] INFO [authorizerStart completed for endpoint PLAINTEXT. Endpoint is now READY. (org.apache.kafka.server.network.EndpointReadyFuture)
[2026-05-26 15:34:24,516] INFO [SocketServer listenerType=BROKER, nodeId=1] Enabling request processing. (kafka.network.SocketServer)
[2026-05-26 15:34:24,517] INFO [Awaiting socket connections on 0.0.0.0:9092. (kafka.network.DataPlaneAcceptor)
[2026-05-26 15:34:24,519] INFO [BrokerServer id=1] Waiting for all of the authorizer futures to be completed (kafka.server.BrokerServer)
[2026-05-26 15:34:24,519] INFO [BrokerServer id=1] Finished waiting for all of the authorizer futures to be completed (kafka.server.BrokerServer)
[2026-05-26 15:34:24,519] INFO [BrokerServer id=1] Waiting for all of the SocketServer Acceptors to be started (kafka.server.BrokerServer)
[2026-05-26 15:34:24,520] INFO [BrokerServer id=1] Finished waiting for all of the SocketServer Acceptors to be started (kafka.server.BrokerServer)
[2026-05-26 15:34:24,521] INFO [BrokerServer id=1] Transition from STARTING to STARTED (kafka.server.BrokerServer)
[2026-05-26 15:34:24,521] INFO [Kafka version: 4.2.0 (org.apache.kafka.common.utils.AppInfoParser)
[2026-05-26 15:34:24,522] INFO [Kafka commitId: a18251bae0b825c6 (org.apache.kafka.common.utils.AppInfoParser)
[2026-05-26 15:34:24,522] INFO [Kafka startTimeMs: 1779824064521 (org.apache.kafka.common.utils.AppInfoParser)
[2026-05-26 15:34:24,525] INFO [KafkaRaftServer nodeId=1] Kafka Server started (kafka.server.KafkaRaftServer)
[2026-05-26 15:44:14,297] INFO [NodeToControllerChannelManager id=1 name=registration] Node 1 disconnected. (org.apache.kafka.clients.NetworkClient)
```

```
[2026-04-22 18:14:19,898] INFO [Awaiting socket connections on 0.0.0.0:9092. (kafka.network.DataPlaneAcceptor)
[2026-04-22 18:14:19,898] INFO [BrokerServer id=1] Waiting for all of the authorizer futures to be completed (kafka.server.BrokerServer)
[2026-04-22 18:14:19,899] INFO [BrokerServer id=1] Finished waiting for all of the authorizer futures to be completed (kafka.server.BrokerServer)
[2026-04-22 18:14:19,899] INFO [BrokerServer id=1] Waiting for all of the SocketServer Acceptors to be started (kafka.server.BrokerServer)
[2026-04-22 18:14:19,899] INFO [BrokerServer id=1] Finished waiting for all of the SocketServer Acceptors to be started (kafka.server.BrokerServer)
[2026-04-22 18:14:19,899] INFO [BrokerServer id=1] Transition from STARTING to STARTED (kafka.server.BrokerServer)
[2026-04-22 18:14:19,900] INFO [Kafka version: 4.2.0 (org.apache.kafka.common.utils.AppInfoParser)
[2026-04-22 18:14:19,900] INFO [Kafka commitId: a18251bae0b825c6 (org.apache.kafka.common.utils.AppInfoParser)
[2026-04-22 18:14:19,900] INFO [Kafka startTimeMs: 1776896059899 (org.apache.kafka.common.utils.AppInfoParser)
[2026-04-22 18:14:19,903] INFO [KafkaRaftServer nodeId=1] Kafka Server started (kafka.server.KafkaRaftServer)
```

In case of errors: create config/log4j2.properties file

```
$content = @"
```

```
status = error
```

```
name = KafkaConfig
```

```
packages = org.apache.kafka
```

```
appender.console.type = Console
```

```
appender.console.name = STDOUT
```

```
appender.console.layout.type = PatternLayout
```

```
appender.console.layout.pattern = [%-5level] %d{yyyy-MM-dd HH:mm:ss.SSS} [%t] %c{1} - %msg%n
```

```
rootLogger.level = INFO
```

```
rootLogger.appenderRefs = stdout
```

```
rootLogger.appenderRef.stdout.ref = STDOUT
```

```
logger.kafka.name = kafka  
logger.kafka.level = INFO  
logger.kafka.appenderRefs = stdout  
logger.kafka.appenderRef.stdout.ref = STDOUT
```

```
logger.zookeeper.name = org.apache.zookeeper  
logger.zookeeper.level = ERROR  
logger.zookeeper.appenderRefs = stdout  
logger.zookeeper.appenderRef.stdout.ref = STDOUT
```

```
logger.kafka.authorizer.name = kafka.authorizer  
logger.kafka.authorizer.level = INFO  
logger.kafka.authorizer.appenderRefs = stdout  
logger.kafka.authorizer.appenderRef.stdout.ref = STDOUT
```

```
logger.kafka.log4j.name = kafka.log4j  
logger.kafka.log4j.level = INFO  
logger.kafka.log4j.appenderRefs = stdout  
logger.kafka.log4j.appenderRef.stdout.ref = STDOUT
```

```
logger.kafka.request.name = kafka.request  
logger.kafka.request.level = WARN  
logger.kafka.request.appenderRefs = stdout  
logger.kafka.request.appenderRef.stdout.ref = STDOUT
```

```
logger.jetty.name = org.eclipse.jetty  
logger.jetty.level = INFO  
logger.jetty.appenderRefs = stdout  
logger.jetty.appenderRef.stdout.ref = STDOUT
```

```
logger.state.change.name = state.change  
logger.state.change.level = INFO  
logger.state.change.appenderRefs = stdout  
logger.state.change.appenderRef.stdout.ref = STDOUT
```

```
"@
```

```
Set-Content -Path "config\log4j2.properties" -Value $content -Encoding UTF8
```

Similarly, modify server.properties
Add the 2 lines

```
node.id=1  
controller.quorum.voters=1@localhost:9093
```

Create topic

```
bin\windows\kafka-topics.bat --create --topic payment-events --bootstrap-server localhost:9092 --partitions 1 --replication-factor 1
```

```
PS C:\Users\saito\Downloads\kafka> bin\windows\kafka-topics.bat --create --topic payment-events --bootstrap-server localhost:9092 --partitions 1 --replication-factor 1
2026-05-26T21:44:33.691418200Z main ERROR Reconfiguration failed: No configuration found for '764c12b6' at 'null' in 'null'
Created topic payment-events.
```

```
PS C:\Users\saito\Downloads\kafka> bin\windows\kafka-topics.bat --list --bootstrap-server localhost:9092
2026-05-26T21:45:06.685733100Z main ERROR Reconfiguration failed: No configuration found for '764c12b6' at 'null' in 'null'
payment-events
```

Powershell

Invoke-WebRequest -Uri "http://localhost:8080/payments/send" -Method POST

or

curl.exe -X POST <http://localhost:8080/payments/send>

From Powershell

```
PS C:\Users\saito\Downloads\kafka>
PS C:\Users\saito\Downloads\kafka> curl.exe -X POST http://localhost:8080/payments/send
Payment event sent to Kafka
```

Consumer shows on the Console Log

```
r clientId=consumer-payment-notification-group-1, groupId=payment-notification-group] Adding newly assigned partitions: [payment-event
s-0]
2026-05-26T17:50:44.653-04:00 INFO 221808 --- [KafkaAnnotation] [ntainer#0-0-C-1] o.a.k.c.c.internals.ConsumerCoordinator : [Consume
r clientId=consumer-payment-notification-group-1, groupId=payment-notification-group] Found no committed offset for partition payment-
events-0
2026-05-26T17:50:44.719-04:00 INFO 221808 --- [KafkaAnnotation] [ntainer#0-0-C-1] o.a.k.c.c.internals.SubscriptionState : [Consume
r clientId=consumer-payment-notification-group-1, groupId=payment-notification-group] Resetting offset for partition payment-events-0
to position FetchPosition{offset=0, offsetEpoch=Optional.empty, currentLeader=LeaderAndEpoch{leader=Optional[localhost:9092 (id: 1 rac
k: null isFenced: false)], epoch=0}}.
2026-05-26T17:50:44.722-04:00 INFO 221808 --- [KafkaAnnotation] [ntainer#0-0-C-1] o.s.k.l.KafkaMessageListenerContainer : payment-
notification-group: partitions assigned: [payment-events-0]
Received payment event from Kafka:
Payment ID: ea3683b9-c275-4e8c-9f56-7532852703e9
Customer ID: CUST001
Amount: 100.0
Status: COMPLETED
Notification sent to customer.
[]
```